ARM Physical IP, Inc.

TSMC 90nm Enhanced (CLN90G) Process, Low-K Dielectric
1.0V SAGE-X(tm) v3.0 Standard Cell Library Release Notes

2005q3v2 (09/02/2005)

Impact on Designs Using 2005q3v1

This section describes the impact of this release on current designs
which are using the previous release.

*** This is a mandatory update. ***

ARM Update Recommendation:

Install version 2005q3v2. Discontinue using 2005q3v1.

o The following cells have changed P&R footprint and gds2 only (no change to timing models) to update these cells to DRC version 1.4 rules and subset DRC rule PO.S.10.1:
```
ACCSIHCONX2 ACCSIHCONX4  ACHCINX2     ACHCONX2     AFCSHCINX2
AFCSHCINX4  AHHCONX2     AND3X4       AND3X8       AND4X6
AOI21X2     AOI2BB2X4    AOI31X2      AOI32X2      AOI33X4
CLKBUFX12   CLKBUFX16    CLKBUFX20    CLKBUFX24    CLKBUFX32
CLKBUFX40   CLKBUFX6     CLKBUFX8     CLKINVX12    CLKINVX16
CLKINVX20   CLKINVX24    CLKINVX32    CLKINVX3     CLKINVX40
CLKINVX6    CLKMX2X12    CLKMX2X3     CLKMX2X4     CLKNAND2X4
CLKNAND2X8  DFFHQX2      DFFHQX4      DFFHQX8      DFFSQX1
DFFSQX2     DFFSQX4      DFFSQXL      DLY1X4       DLY2X4
DLY3X4      DLY4X4       EDFFHQX4     EDFFHQX8     MDFFHQX2
MXI2X2      MXI2X6       NAND2X6      NAND2X8      NAND3X2
NAND3X4     NAND3X6      NAND3X8      NAND4X2      NAND4X4
NAND4X6     NAND4X8      NOR2BX8      NOR2X4       NOR2X8
NOR3X2      NOR3X8       NOR4X8       OA21X2       OA21X4
OAI211X2    OAI211X4     OAI21X2      OAI22X2      OAI2B2X4
OAI31X2     OAI31X4      OAI32X2      OAI32X4      OAI33X4
OR3X4       OR4X4        SDFFHQX8     SDFFNSRHX8   SDFFSHQX4
SDFFSHQX8   SDFFSQX1     SDFFSQX2     SDFFSQX4     SDFFSQXL
SEDFFHQX4   SEDFFHQX8    TLATNCAX12   TLATNCAX16   TLATNCAX20
TLATNCAX8   TLATNTSCAX12 TLATNTSCAX16 TLATNTSCAX20 TLATNTSCAX4
TLATNTSCAX8 TLATSRX4
```
o The following cells have changed gds2 to add dummy poly (DPO):
  FILL8, FILL16, FILL32, FILL64
o Timing and power have not changed.
o Astro tf and library updated to DRC 1.4.
o LEF technology headers updated to DRC 1.4.
o The process of updating our product development infrastructure to change our new corporate name from "Artisan Components, Inc." to "ARM Physical IP, Inc." is in progress, and this product release reflects this change. You may still find instances of the company name "Artisan Components, Inc." or "Artisan" in these product files. This is not an error but is an artifact of the logistical difficulties in updating all aspects of our product development infrastructure simultaneously. Note that the Artisan brand name remains a legitimate product description, and continues to be used where appropriate.

```
Process Technology Data
-----------------------
This section lists all of the foundry documents and technology files used to
design this product.

  o Design Rule Document
      - T-N90-LO-DR-001 Rev 1.4 06-09-05
  o DRC Command File
      - CALIBRE DRC COMMAND FILE CLN90S_9M.14a (Version 1.4a 06-01-2005)
      - CALIBRE DRC COMMAND FILE CLN80S_9M.02a (Version 0.2a 07-22-2005)
        - CLN90S_PO_ISO.13a.eva012805 v1.3a 1/28/2005
  o LVS Command File
      - CALIBRE LVS COMMAND FILE LVS_RC_Calibre_90nm_combo_deck_1p9m+RDL.13a
        (Version 1.3a 01-04-2005)
  o Process Spice Models
      - T-N90-LO-SP-002 Version 1.1A December 23rd, 2003
  o Extractor Technology File
      - T-N90-LO-SP-002-B1 v-prelim received on 02/23/2004

EDA Support
-----------
This section lists the EDA tools and versions supported for this
product release. The items indicated by the "*" were used during the
QA testing of this release. This set of tools and versions corresponds
to ARM EDA Package 3.1-90nm.

    * Cadence NC-Sim (Verilog)
        - 3.4
    * Synopsys VCS
        - 6.2
    o Mentor ModelSim (Verilog)
        - 5.7d
    o Cadence NC-Sim (VHDL)
        - 3.4
    * Mentor ModelSim (VHDL)
        - 5.7d
    * Simulation Model SDF Compatibility
        - SDF 2.1
    * Synopsys Design Compiler (Synthesis, Delay Calculation)
        - 2002.05
    o Synopsys Module Compiler
        - 2002.05
    * Cadence BuildGates/PKS with .lib
        - 5.08
    * Synopsys Physical Compiler
        - 2002.05
    * Synopsys PrimeTime (Static Timing, Delay Calculation)
        - 2002.03-SP1
    o Synopsys Floorplan Compiler
        - 2002.05
    * Cadence NanoRoute/SoC
        - 2.1/3.3
    o Synopsys Astro (FRAM, CEL, .lib/.db)
        - 2003.09
    * Mentor Graphics Calibre (GDSII, CDL)
```

```
        - 2003.2_12
    * Cadence Composer (Schematic Entry)
        - DFII 4.4.5
    * Synopsys Power Compiler
        - 2002.05
    o Synopsys Prime Power
        - 2002.05
    * Synopsys PrimeTime SI
        - 2002.03-SP1
    o Synopsys DFT Compiler
        - 2002.05
    * Adobe Acrobat Reader (PDF Documentation)
        - 5.0
    * Cadence Spectre
        - 5.0.33.011504

Technology Implementation
-------------------------
This section provides information on items that may not be included in the
foundry documentation.

    Characterization Corners:

      fast            : FF Process Spice Models, 1.1V, -40 Degrees C.
      fastz           : FF Process Spice Models, 1.1V,   0 Degrees C.
      fast_leakage    : FF Process Spice Models, 1.1V, 125 Degrees C.
      typical         : TT Process Spice Models, 1.0V,  25 Degrees C.
      typical_leakage : TT Process Spice Models, 1.0V,  50 Degrees C.
      slow            : SS Process Spice Models, 0.9V, 125 Degrees C.

  Mandatory Tapeout layers required for foundry:

      - GDS2 text layer 63:63 is allocated by the foundry for IP marking.
        *** This is a mandatory tapeout layer ****

Methodology
-----------
This section describes design flow issues that may be applicable to this
release.

  o DRC
      - The FILL1 cell will fail a variety of DRC errors when run on
        FILL1 individually. These errors will disappear at chip-level DRC
        due to interaction with neighboring cells.

  o LVS
      - 79 cells will have LVS violations when LVS is run on the cells in
        isolation because there is no direct well connection. All LVS errors
        will disappear when each cell is abutted next to another cell as
        in a real design.

      - The following cells will have "isolated layout nets" due to PO
        added to meet PO.S.10.1 rule:

        MDFFHQX2      MXI2X6       NAND2X8  NAND3X4  NAND3X6  NAND3X8
        NAND4X4       NOR3X2       NOR3X8   NOR4X8   TLATNCAX20
        TLATNTSCAX16  TLATNTSCAX20
```

```
  o ERC
      - The following cells will have ERC violations when LVS is run
        at the cell level because there is no power/ground path to
        the gates. All ERC errors will disappear when another cell is driving
        these cells as in a real design.

        MXI2DXL   MXI2DX1     MXI2DX2     MXI2DX4

      - The following cells will have ERC violations when you run LVS
        because the gate of the transistors doesn't have path to power/ground.
        The user has to add these cells manually into the design netlist to
        pass the LVS as these cells have devices in them.

        FILLCAP3   FILLCAP4   FILLCAP8   FILLCAP16   FILLCAP32   FILLCAP64

  o The TLATNCA* and TLATNTSCA* cells are not compatible with the
    SDF writer in PrimeTime version 2002.03-SP1 and will result in
    SDF files that will not back-annotate correctly with ARM
    timing simulation models. The problem can be worked around by
    removing this line from all TLATN*CA* instances in the SDF:

    (IOPATH CK CKcheckpin1 (0.000:0.000:0.000))

    You should also delete "checkpin1" from the SETUP and HOLD checks in
    the SDF file for all TLATN*CA* cells.

Known Problems and Solutions
----------------------------
This section describes known problems and work-arounds as of the
release date of this library.

  o Place and route tools may create M1.S.2 drc violations when dropping
    metal1 vias to standard cell pins. The above versions (see EDA Views
    section) of place and route tools cannot always avoid these violations.
    This issue is under investigation.

  o The sdf from BuildGates for the following cells contain
    an RN -> Q arc with no data in it. ARM is currently investigating
    this with Cadence:

    SDFFRQXL  SDFFRQX1  SDFFRQX2  SDFFRQX4
    SDFFRHQX1 SDFFRHQX2 SDFFRHQX4 SDFFRHQX8

-------------------------------------------------------------------------------
2005q3v1 (07/28/2005)

Impact on Designs Using 2004q2v2
--------------------------------
This section describes the impact of this release on current designs
which are using the previous release.

*** This is a mandatory update. ***

ARM Update Recommendation:

  Install version 2005q3v1. Discontinue using 2004q2v2.
```

o The following cells have changed P&R footprint and gds2:
CLKBUFX6 DFFSQX1 DFFSQX2 DFFSQX4 DFFSQXL
SDFFSHQX4 SDFFSHQX8 SDFFSQX1 SDFFSQX2 SDFFSQX4 SDFFSQXL
TLATNCAX12 TLATNTSCAX12
These cells have been modified to improve reliability due to electromigration. Timing values for these cells have not changed.
Prior layout revisions are still safe to manufacture; however, the new layouts will have improved long term reliability when exposed to extremely high toggle rates and high loading conditions for long periods of time.

o All views have the following cells removed because busholder cells have been found impractical for 90nm designs:
HOLDX1 RFRDX1 RFRDX2 RFRDX4

o Timing has changed to remove the 4 cells listed above.

o Timing has changed for the SEDFF* cells. The setup and hold windows for these cells have increased.

o Timing has changed for the TLATN*C* cells. The setup and hold windows for these cells have increased to reduce the CK-to-ECK delay tolerance to 2%.

o Updated Astro tf to update to DRC 1.3a, to change the power units mW to pW in astro techfiles, add layers VTLN and VTLP (for compatibility with LVt library), and to remove maskName definition from layer ARTISCANTEXT. Loaded tf into Astro library and reloaded CLF files.

o Astro antenna clf updated to handle both accumulate-layer and single-layer ratio rules simultaneously.

o LEF header modified to add density checks and to update to DRC 1.3a.

o The process of updating our product development infrastructure to change our new corporate name from "Artisan Components, Inc." to "ARM Physical IP, Inc." is in progress, and this product release reflects this change. You may still find instances of the company name "Artisan Components, Inc." or "Artisan" in these product files. This is not an error but is an artifact of the logistical difficulties in updating all aspects of our product development infrastructure simultaneously. Note that the Artisan brand name remains a legitimate product description, and continues to be used where appropriate.

Process Technology Data

This section lists all of the foundry documents and technology files used to design this product.

o Design Rule Document
  - T-N90-LO-DR-001 Rev 1.1 12-05-03

o DRC Command File
  - CALIBRE DRC COMMAND FILE CLN90S_9M.11a.2 (Version 1.1a 12-01-2003)

o LVS Command File
  - CALIBRE LVS COMMAND FILE LVS_RC_Calibre_90nm_combo_deck.10a (Version 1.0a 10-02-2003)

o Process Spice Models
  - T-N90-LO-SP-002 Version 1.1A December 23rd, 2003

o Extractor Technology File
  - T-N90-LO-SP-002-B1 v-prelim received on 02/23/2004

EDA Support

This section lists the EDA tools and versions supported for this product release. The items indicated by the "*" were used during the QA testing of this release. This set of tools and versions corresponds to ARM EDA Package 3.1-90nm.

```
    * Cadence NC-Sim (Verilog)
        - 3.4
    * Synopsys VCS
        - 6.2
    o Mentor ModelSim (Verilog)
        - 5.7d
    o Cadence NC-Sim (VHDL)
        - 3.4
    * Mentor ModelSim (VHDL)
        - 5.7d
    * Simulation Model SDF Compatibility
        - SDF 2.1
    * Synopsys Design Compiler (Synthesis, Delay Calculation)
        - 2002.05
    o Synopsys Module Compiler
        - 2002.05
    * Cadence BuildGates/PKS with .lib
        - 5.08
    * Synopsys Physical Compiler
        - 2002.05
    * Synopsys PrimeTime (Static Timing, Delay Calculation)
        - 2002.03-SP1
    o Synopsys Floorplan Compiler
        - 2002.05
    * Cadence NanoRoute/SoC
        - 2.1/3.3
    o Synopsys Astro (FRAM, CEL, .lib/.db)
        - 2003.09
    * Mentor Graphics Calibre (GDSII, CDL)
        - 2003.2_12
    * Cadence Composer (Schematic Entry)
        - DFII 4.4.5
    * Synopsys Power Compiler
        - 2002.05
    o Synopsys Prime Power
        - 2002.05
    * Synopsys PrimeTime SI
        - 2002.03-SP1
    o Synopsys DFT Compiler
        - 2002.05
    * Adobe Acrobat Reader (PDF Documentation)
        - 5.0
    * Cadence Spectre
        - 5.0.33.011504
```

Technology Implementation

This section provides information on items that may not be included in the foundry documentation.

Characterization Corners:

```
fast             : FF Process Spice Models, 1.1V, -40 Degrees C.
fastz            : FF Process Spice Models, 1.1V,   0 Degrees C.
fast_leakage     : FF Process Spice Models, 1.1V, 125 Degrees C.
typical          : TT Process Spice Models, 1.0V,  25 Degrees C.
typical_leakage  : TT Process Spice Models, 1.0V,  50 Degrees C.
slow             : SS Process Spice Models, 0.9V, 125 Degrees C.
```

Mandatory Tapeout layers required for foundry:

- GDS2 text layer 63:63 is allocated by the foundry for IP marking.
  *** This is a mandatory tapeout layer ****

Methodology

This section describes design flow issues that may be applicable to this release.

o DRC
- The FILL1 cell will fail a variety of DRC errors when run on FILL1 individually. These errors will disappear at chip-level DRC due to interaction with neighboring cells.

o LVS
- 79 cells will have LVS violations when LVS is run on the cells in isolation because there is no direct well connection. All LVS errors will disappear when each cell is abutted next to another cell as in a real design.

o ERC
- The following cells will have ERC violations when LVS is run at the cell level because there is no power/ground path to the gates. All ERC errors will disappear when another cell is driving these cells as in a real design.

  MXI2DXL   MXI2DX1   MXI2DX2   MXI2DX4

- The following cells will have ERC violations when you run LVS because the gate of the transistors doesn't have path to power/ground. The user has to add these cells manually into the design netlist to pass the LVS as these cells have devices in them.

  FILLCAP3   FILLCAP4   FILLCAP8   FILLCAP16   FILLCAP32   FILLCAP64

o The TLATNCA* and TLATNTSCA* cells are not compatible with the SDF writer in PrimeTime version 2002.03-SP1 and will result in SDF files that will not back-annotate correctly with ARM timing simulation models. The problem can be worked around by removing this line from all TLATN*CA* instances in the SDF:

```
(IOPATH CK CKcheckpin1 (0.000:0.000:0.000))
```

You should also delete "checkpin1" from the SETUP and HOLD checks in the SDF file for all TLATN*CA* cells.

Known Problems and Solutions

This section describes known problems and work-arounds as of the

release date of this library.

  o Place and route tools may create M1.S.2 drc violations when dropping
    metal1 vias to standard cell pins. The above versions (see EDA Views
    section) of place and route tools cannot always avoid these violations.
    This issue is under investigation.

  o The sdf from BuildGates for the following cells contain
    an RN -> Q arc with no data in it. ARM is currently investigating
    this with Cadence:

    SDFFRQXL SDFFRQX1 SDFFRQX2 SDFFRQX4
    SDFFRHQX1 SDFFRHQX2 SDFFRHQX4 SDFFRHQX8

2004q2v2 (06/03/2004)

Impact on Designs Using 2004q2v1

This section describes the impact of this release on current designs
which are using the previous release.

*** This is an optional update. ***

Artisan Update Recommendation:

  Install version 2004q2v2 to use new sdf post processor for PrimeTime.
  Discontinue using 2004q2v1.

  o Updated sdf post processor for PrimeTime.

Process Technology Data

This section lists all of the foundry documents and technology files used to
design this product.

  o Design Rule Document
      - T-N90-LO-DR-001 Rev 1.1 12-05-03
  o DRC Command File
      - CALIBRE DRC COMMAND FILE CLN90S_9M.11a.2 (Version 1.1a 12-01-2003)
  o LVS Command File
      - CALIBRE LVS COMMAND FILE LVS_RC_Calibre_90nm_combo_deck.10a
        (Version 1.0a 10-02-2003)
  o Process Spice Models
      - T-N90-LO-SP-002 Version 1.1A December 23rd, 2003
  o Extractor Technology File
      - T-N90-LO-SP-002-B1 v-prelim received on 02/23/2004

EDA Views

This section lists the EDA tools and versions used to verify
this product release. This set of tools and versions corresponds to
Artisan EDA Package 3.1-90nm.

  o Cadence NC-Verilog
      - 3.4
  o Synopsys VCS

```
      - 6.2
  o MTI/Mentor ModelSim (VHDL)
      - 5.7d
  o Cadence BuildGates with Low Power Synthesis
      - v5.0.8
  o Synopsys Design Compiler with Power Compiler
      - 2002.05
  o Synopsys Physical Compiler
      - 2002.05
  o Synopsys PrimeTime
      - 2002.03-SP1
  o Cadence NanoRoute (SoC Encounter)
      - 2.1 (3.3)
  o Synopsys Astro
      - 2003.09
  o Adobe Acrobat Reader
      - 4.0
  o Mentor Calibre
      - 2003.2_12
  o Cadence Spectre
      - 5.0.33.011504

Technology Implementation
-------------------------
This section provides information on items that may not be included in the
foundry documentation.

    Characterization Corners:

      fast            : FF Process Spice Models, 1.1V, -40 Degrees C.
      fastz           : FF Process Spice Models, 1.1V,   0 Degrees C.
      fast_leakage    : FF Process Spice Models, 1.1V, 125 Degrees C.
      typical         : TT Process Spice Models, 1.0V,  25 Degrees C.
      typical_leakage : TT Process Spice Models, 1.0V,  50 Degrees C.
      slow            : SS Process Spice Models, 0.9V, 125 Degrees C.

  Mandatory Tapeout layers required for foundry:

      - GDS2 text layer 63:63 is allocated by the foundry for IP marking.
        *** This is a mandatory tapeout layer ****

Methodology
-----------
This section describes design flow issues that may be applicable to this
release.

  o DRC
      - The FILL1 cell will fail a variety of DRC errors when run on
        FILL1 individually. These errors will disappear at chip-level DRC
        due to interaction with neighboring cells.

  o LVS
      - 79 cells will have LVS violations when LVS is run on the cells in
        isolation because there is no direct well connection. All LVS errors
        will disappear when each cell is abutted next to another cell as
        in a real design.
```

```
  o ERC
      - The following cells will have ERC violations when LVS is run
        at the cell level because there is no power/ground path to
        the gates. All ERC errors will disappear when another cell is driving
        these cells as in a real design.

        MXI2DXL   MXI2DX1     MXI2DX2     MXI2DX4

      - The following cells will have ERC violations when you run LVS
        because the gate of the transistors doesn't have path to power/ground.
        The user has to add these cells manually into the design netlist to
        pass the LVS as these cells have devices in them.

        FILLCAP3   FILLCAP4   FILLCAP8   FILLCAP16   FILLCAP32   FILLCAP64

Known Problems and Solutions
----------------------------
This section describes known problems and work-arounds as of the
release date of this library.

  o Place and route tools may create M1.S.2 drc violations when dropping
    metal1 vias to standard cell pins. The above versions (see EDA Views
    section) of place and route tools cannot always avoid these violations.
    This issue is under investigation.

  o The above version (see EDA Views section) of Astro does not support
    both accumulate-layer and single-layer ratio rules simultaneously.
    Your place and route results may fail the calibre accumulate antenna
    rule A.R.5_A.R.8.MX.  This is a real DRC violation and must be
    addressed manually.

-------------------------------------------------------------------------------
2004q2v1 (04/23/2004)

*** This is an initial release. ***

Process Technology Data
-----------------------
This section lists all of the foundry documents and technology files used to
design this product.

  o Design Rule Document
      - T-N90-LO-DR-001 Rev 1.1 12-05-03
  o DRC Command File
      - CALIBRE DRC COMMAND FILE CLN90S_9M.11a.2 (Version 1.1a 12-01-2003)
  o LVS Command File
      - CALIBRE LVS COMMAND FILE LVS_RC_Calibre_90nm_combo_deck.10a
        (Version 1.0a 10-02-2003)
  o Process Spice Models
      - T-N90-LO-SP-002 Version 1.1A December 23rd, 2003
  o Extractor Technology File
      - T-N90-LO-SP-002-B1 v-prelim received on 02/23/2004

EDA Views
---------
This section lists the EDA tools and versions used to verify
this product release. This set of tools and versions corresponds to
```

Artisan EDA Package 3.1-90nm.

- o Cadence NC-Verilog
  - 3.4
- o Synopsys VCS
  - 6.2
- o MTI/Mentor ModelSim (VHDL)
  - 5.7d
- o Cadence BuildGates with Low Power Synthesis
  - v5.0.8
- o Synopsys Design Compiler with Power Compiler
  - 2002.05
- o Synopsys Physical Compiler
  - 2002.05
- o Synopsys PrimeTime
  - 2002.03-SP1
- o Cadence NanoRoute (SoC Encounter)
  - 2.1 (3.3)
- o Synopsys Astro
  - 2003.09
- o Adobe Acrobat Reader
  - 4.0
- o Mentor Calibre
  - 2003.2_12
- o Cadence Spectre
  - 5.0.33.011504

Technology Implementation

This section provides information on items that may not be included in the foundry documentation.

Characterization Corners:

```
fast            : FF Process Spice Models, 1.1V, -40 Degrees C.
fastz           : FF Process Spice Models, 1.1V,   0 Degrees C.
fast_leakage    : FF Process Spice Models, 1.1V, 125 Degrees C.
typical         : TT Process Spice Models, 1.0V,  25 Degrees C.
typical_leakage : TT Process Spice Models, 1.0V,  50 Degrees C.
slow            : SS Process Spice Models, 0.9V, 125 Degrees C.
```

Mandatory Tapeout layers required for foundry:

- GDS2 text layer 63:63 is allocated by the foundry for IP marking.
  *** This is a mandatory tapeout layer ****

Methodology

This section describes design flow issues that may be applicable to this release.

- o DRC
  - The FILL1 cell will fail a variety of DRC errors when run on FILL1 individually. These errors will disappear at chip-level DRC due to interaction with neighboring cells.

- o LVS

- 79 cells will have LVS violations when LVS is run on the cells in isolation because there is no direct well connection. All LVS errors will disappear when each cell is abutted next to another cell as in a real design.

o ERC

- The following cells will have ERC violations when LVS is run at the cell level because there is no power/ground path to the gates. All ERC errors will disappear when another cell is driving these cells as in a real design.

  MXI2DXL   MXI2DX1   MXI2DX2   MXI2DX4

- The following cells will have ERC violations when you run LVS because the gate of the transistors doesn't have path to power/ground. The user has to add these cells manually into the design netlist to pass the LVS as these cells have devices in them.

  FILLCAP3   FILLCAP4   FILLCAP8   FILLCAP16   FILLCAP32   FILLCAP64

Known Problems and Solutions

This section describes known problems and work-arounds as of the release date of this library.

o Place and route tools may create M1.S.2 drc violations when dropping metal1 vias to standard cell pins. The above versions (see EDA Views section) of place and route tools cannot always avoid these violations. This issue is under investigation.

o The above version (see EDA Views section) of Astro does not support both accumulate-layer and single-layer ratio rules simultaneously. Your place and route results may fail the calibre accumulate antenna rule A.R.5_A.R.8.MX. This is a real DRC violation and must be addressed manually.
